SS-P-166d

MARCH 10, 1961

SUPERSEDING
Int. Fed. Spec. SS–P–00166c(GSA–FSS)
October 14, 1960 and
Fed. Spec. SS–P–166a
February 24, 1956

FEDERAL SPECIFICATION

PENCILS, LEAD

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers wood-cased lead pencils primarily for use as a marking medium on lumber and paper.

1.2 Classification.

1.2.1 *Types, classes, and degrees of hardness.*—The pencils shall be of the following types, classes, and degrees of hardness, as specified (see table I and par. 6.2):

Type I.—Carpenter.

Type II.—Checking.

Type III.—Copying.
- Class 1.—Plain (without mouthpiece or point protector).
- Class 2.—With mouthpiece only.
- Class 3.—With point protector only.
- Class 4.—With mouthpiece and point protector.
- Medium lead.
- Hard lead.

Type IV.—General writing.
- No. 1.—Soft lead.
- No. 2.—Medium lead.
- No. 3.—Medium hard lead.

Type V.—Extra thin lead.
- No. 2.—Medium soft.
- No. 2½—Firm.
- No. 3.—Hard.

Type VI.—Stenographer.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:
- GG–S–236—Sharpeners, Pencil.
- UU–P–121—Paper, Bond and Writing, White and Colored.
- ZZ–E–661—Erasers; Rubber and Rubber Substitutes.
- PPP–B–636—Boxes, Fiberboard.
- PPP–T–97 — Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

Federal Standards:
- Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
- Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, At-

570403/61/270

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

lanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL–L–10547 — Liners, Case, Waterproof.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Lead.*—The lead shall be made from a black graphite mixture and a graphite and dye composition for copying pencil (type III), and shall produce smooth, even, and uniform writing (see 3.5.3 and 4.4.1).

3.1.2 *Wood.*—Each casing shall be made of soft, well-seasoned, straight-grained wood, shall not split or splinter, and shall be readily cut with a pencil sharpener conforming to type II of Federal Specification GG–S–236. The wood shall take a smooth, even finish.

3.2 Construction (general).—Pencils shall be straight and both halves of the wood casing shall be of uniform structure, securely glued together (bonded) the entire length. The lead shall be in one piece and shall be enclosed in the wood casing in a neat, secure manner and as specified hereinafter. The lead shall be centered within a tolerance of 0.01 inch the entire length of the pencil.

3.3 Leads.—Pencils shall be provided with leads of the dimensions or diameters, and degree of hardness for the respective types as indicated in table I.

TABLE I.

Type	Degree of hardness (lead)	Dimensions or diameters of leads (inch)	
		Minimum	Maximum
I, Carpenter	Hard	0.0625 (thick) .1875 (wide)	--- ---
II, Checking	Soft	.125	0.135
III, Copying	Medium Hard	.095 .090	.110 .110
IV, General writing	No. 1, soft No. 2, medium No. 3, medium hard	.093 .081 .075	.115 .095 .083
V, Extra thin lead	No. 2, medium soft [1]No. 2½, firm No. 3, hard	.065 .058 .058	.073 .072 .071
VI, Stenographer	[1]Medium soft	.080	.092

[1] Other comparable manufacturers' identifications are acceptable.

3.4 Identification marking.—Each pencil shall be permanently marked with the name of the manufacturer. Types III, IV, and V pencils shall be permanently marked with the degree of hardness. Types II, IV and V pencils shall be permanently marked with information that the lead is bonded or fused to the wood casing. Additional identifying markings consistent with commercial practice may be imprinted, and for this purpose blind stamping is permissible.

3.5 Types.

3.5.1 *Type I, carpenter.*—Carpenter pencils shall have lead as indicated in table I. The casing shall be oval or flat sided in shape and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

shall be finished with bright colored lacquer. The length of the finished pencil shall be not less than 6¾ inches.

3.5.2 *Type II, checking.*—Checking pencils shall have lead as indicated in table I. The pencils shall be untipped, round or hexagonal and shall be finished with a good polish; the color to approximate the color of the lead. Pencils shall be not less than 6¾ inches long. Lead shall be bonded to the wood casing.

3.5.3 *Type III, copying.*—Copying pencils shall have lead as indicated in table I and shall be provided with medium or hard lead, as specified (see 1.2.1 and 6.2). The lead shall be indelible and made from a graphite and dye composition. The pencil shall write black and copy purple. Casings shall be round and finished with a good polish, natural or in colors. Pencils shall be not less than 6¾ inches long furnished in the classes specified in 1.2.1 (see 6.2). Mouthpieces and point protectors, when required (see 1.2.1), shall be suitable for the purpose intended and shall be in accordance with the quality requirements of this specification.

3.5.4 *Type IV, general writing.*—General writing pencils shall be not less than 6 1/16 inches long, not including tip and eraser. Each pencil shall be provided with a suitable metal tip which firmly secures a rubber eraser. The eraser shall be in accordance with Federal Specification ZZ-E-661, type I, grade B. Pencils shall be round or hexagonal and shall be finished with a colored lacquer. Leads shall be bonded to the wood casing and shall not crumble or break under ordinary use. The lead hardness shall be as specified (see 1.2.1 and 6.2) and the lead diameter shall be as indicated in table I.

3.5.5 *Type V, extra thin lead.*—Extra thin lead pencils shall be constructed and finished in accordance with the requirements for type IV, general writing pencils. The lead hardness shall be as specified (see 1.2.1 and 6.2) and the lead diameter shall be as indicated in table I.

3.5.6 *Type VI, stenographer.*—Stenographer pencils shall be untipped and sharpened at both ends. The casing shall be round, ¼ inch ± 1/32 inch in diameter and finished with a good polish, natural or in colors. The lead shall be medium soft, and the lead diameter shall be as indicated in table I. The pencil shall have a minimum length of 6⅞ inches.

3.6 **Workmanship.**—Lead pencils shall be smoothly finished and shall be free from splits, splinters, blisters and any other imperfections which affect the appearance or which might impair the serviceability. Mouthpieces and protectors, when required with type III copying pencils, shall be free from imperfections which affect the appearance or usefulness of these items.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier's responsibility.** — Unless otherwise specified (see 6.2), supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Preservation, packaging, packing and marking of pencils shall be subject to inspection and test by the Government inspector to determine compliance with the requirements of section 5 of this specification.

4.2 **Sampling for lot acceptance.**

4.2.1 *Inspection lot.*—A lot shall consist of all pencils of the same type, class, and degree of hardness offered for delivery at one time. The unit of product shall be a package of 1 dozen pencils.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

4.2.2 *Sampling for examination.* — The sample shall consist of one pencil from each unit of product selected in accordance with Inspection Level L4 of Military Standard MIL–STD–105.

4.2.3 *Sampling for test.*—The sample shall consist of one pencil from each unit of product selected in accordance with Inspection Level L2 of MIL–STD–105.

4.3 **Examination.** — Each of the sample pencils taken in accordance with 4.2.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table II. The Acceptable Quality Level (AQL) shall be 2.5 percent defective for major defects.

TABLE II.—*Classification of defects in accordance with Military Standard MIL–STD–105*

Categories[1]	Defects
Critical:	None defined.
Major:	
101	Wood does not take a smooth, even finish when sharpened.
102	Casing not securely glued together.
103	Lead not centered, within tolerance specified.
104	Length not as specified.
105	Lead not bonded to casing, types II, IV, and V.
106	Eraser loose, types IV and V.
107	Imperfections in finish affecting appearance or serviceability.
108	Identification marking missing or illegible.
109	Mouthpiece or point protector missing, type III.
Minor:	None defined.

[1] Numerals listed adjacent to description of defects are for use of the inspector in recording of defects.

4.4 **Test.**

4.4.1 *Performance.* — Each sample pencil shall be sharpened in a pencil sharpener conforming to type II of Federal Specification GG–S–236 to a fine point at seven different points and tested at each point by drawing a line equivalent to 24 inches under ordinary writing pressure. The line(s) shall show uniformity of color and evenness as required by this specification. When drawing a line, the lead shall work smoothly, without grittiness. Drawing shall be made on type IV, bond paper conforming to Federal Specification UU–P–121, except for type I pencils. Lines shall be drawn with type I pencils on dressed or planed lumber.

5. PREPARATION FOR DELIVERY

5.1 Federal Standard No. 102 (for Civil agencies only).—Application of the various levels of packaging and packing protection shall be as defined in Federal Standard No. 102.

5.2 Packaging (see 6.2).

5.2.1 *Levels A and C (Military agencies).* —Pencils shall be boxed or banded 1 dozen per package in accordance with standard commercial practice. Unless otherwise specified, six unit packages shall be packaged in standard commercial boxes.

5.2.2 *Levels B and C (Civil agencies).*

5.2.2.1 *Unit package.*—Twelve pencils of one type, class, degree of hardness, etc., as applicable, shall be packaged in a paperboard open-end sleeve, clear plastic covered tray, or other suitable container in accordance with the manufacturer's regular commercial practice.

5.2.2.2 *Intermediate package.* — Unless otherwise specified, six unit packages of pencils of one type, class, degree of hardness, etc., as applicable, shall be packaged in a paperboard box or carton of the type regularly used by the manufacturer.

5.3 Packing (see 6.2).

5.3.1 *Level A.*—Pencils shall be packed in overseas-type fiber boxes conforming to Federal Specification PPP–B–636, class 2, V3c or V3s. Box closures, sealing and strapping

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

shall conform to the box specification and the appendix thereto. In lieu of steel strapping, boxes may be reinforced with pressure-sensitive filament tape conforming to type IV of Federal Specification PPP–T–97, applied in accordance with the appendix thereto. Boxes shall be provided with case liners conforming to Military Specification MIL–L–10547, sealed in accordance with the appendix thereto. The quantity per shipping container shall be as specified in the contract or order (see 6.2). In the absence of specified quantities, the gross weight limitations of the box specification shall apply.

5.3.2 *Level B.*—Unless otherwise specified, twenty intermediate boxes or cartons of 6 dozen (6 unit packages) each, a total of 120 dozen pencils of one type, class, degree of hardness, etc., as applicable, shall be packed in a fiberboard shipping container conforming to Federal Specification PPP–B–636, type I or II, class 1, style RSC. Closure shall be in accordance with the appendix thereto.

5.3.3 *Level C.*—Unless otherwise specified, pencils shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.4 Marking and labeling.

5.4.1 *Civil agencies.*—Marking shall be in accordance with Federal Standard No. 123, and as follows:

5.4.1.1 *Unit package.*—Marking in accordance with manufacturer's standard commercial practice will be acceptable.

5.4.1.2 *Intermediate package.* — Unless otherwise specified, each intermediate package shall be marked or labeled on one end with the following information: Federal Stock Number; item identification, including type, class, degree of hardness, etc., as applicable; the quantity contained therein (e.g., 6 Dozen; not ½ Gross); contract number; applicable specification number; and the name and address of the manufacturer or contractor, if different.

5.4.1.3 *Shipping container.*—In addition to any special marking required in the contract or order, each shipping container shall be marked in accordance with 5.4.1, including specification classification of the commodity (type, class, etc.)

5.4.2 *Military agencies.*

5.4.2.1 *Unit package.*—Each unit package shall be marked in accordance with manufacturer's commercial practice.

5.4.2.2 *Intermediate and shipping containers.*—In addition to any special marking required by the contract or order, intermediate and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

5.5 Standard pack (Civil agencies).—The standard pack requirements for the items covered by this specification shall be as specified in 5.2.2, 5.3.2, and 5.4.1 (see 6.4).

6. NOTES

6.1 Intended use.

6.1.1 *Type I.*—Carpenter pencils are intended for marking on dressed or planed lumber.

6.1.2 *Type II.*—Checking pencils are intended for freehand writing and marking, sketching, layouts, proofreading, checking and making bold markings.

6.1.3 *Type III.*—Copying pencils are intended for use in duplicator and hectograph work and for writing on legal documents or papers where erasures are forbidden or undesirable.

6.1.4 *Type IV.*—General writing pencils are intended for writing notes, memoranda,

Source: https://assist.dla.mil -- Downloaded: 2026-04-26T12:12Z
Check the source to verify that this is the current version before use.

rough drafts, and other general writing purposes.

6.1.5 *Type V.*—Extra-thin lead pencils are intended for fine line writing and for use by accountants and statisticians; and for use by draftsmen and artists where use of drawing pencils is not essential.

6.1.6 *Type VI.*—Stenographer pencils are intended for writing shorthand and other pencil notes by stenographers.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein (see 5.2.2.2, 5.3.2, 5.3.3, and 5.4.1.2), and procurement documents should specify the following (see 6.3 and 6.4):

(*a*) Title, number, and date of this specification.
(*b*) Type of pencil desired (see 1.2.1).
(*c*) Class when type II pencils are required (see 1.2.1).
(*d*) Degree of lead hardness when type III, IV or V pencils are required (see 1.2.1).
(*e*) Data, as required, regarding examination and tests; and inspection responsibility (for Civil agencies only) if other than as specified herein (see 4.1).
(*f*) Both the level of packaging and the level of packing and any special marking required (see 5.2, 5.3, and 5.4).
(*g*) Quantity per shipping container, level A packing (see 5.3.1).

6.3 Military standards.—Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 Standard pack (civil agencies).—The standard pack requirements in 5.2.2, 5.3.2, and 5.4.1 are intended for use in procurements of Stores Stock replenishments. Procuring officers should use the standard pack provisions when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot or intermediate storage point for temporary storage, subsequent issue and/or shipment to eventual user.

6.5 Transportation description.—Transportation description and minimum weights applicable to this item are:

Pencils, not otherwise indexed by name. Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Truckload minimum weight 24,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:
Army—Quartermaster Corps
Navy—Bureau of Supplies and Accounts
Air Force—Mobile Air Materiel Area.

Copies of this specification may be purchased for 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:12Z
Check the source to verify that this is the current version before use.